

(a) Extreme Zero-Inflation

94.2%

16:1
zero-to-positive
ratio

5.8%

Zero Demand
($y = 0$)

Positive Demand
($y > 0$)

Distribution of Positive Demand

-- Mean: 1.30
... Median: 1.00

Charging Sessions per Hour ($y \mid y > 0$)

(b) Station Lifecycle and Cold-Start Evaluation Window

**Cold-Start Window
(14 days)**

Observation density

*Challenge:
New stations have
limited history for
temporal learning*

Stations accumulate
historical data

Relative Observation Density

Days Since Station Opening